Product Specification

Part Name: TFT Display Module
Customer Part ID:

Customer:
Approved by

Revised History

Part Number	Revision	Revision Content	Revised on
2424KBWPG09	A	New	2021-07-13

Contents

Revision History
Contents
1. General Description
2. Mechanical Drawing
3. Pin Description
4. Electrical Characteristics
5. Optical Characteristics
6. Reliability
7. Package Specifications
8. Incoming Inspection standards
9. Precautions When Using These TFT Display Modules
10. Warranty
11. Notice

1. General Description

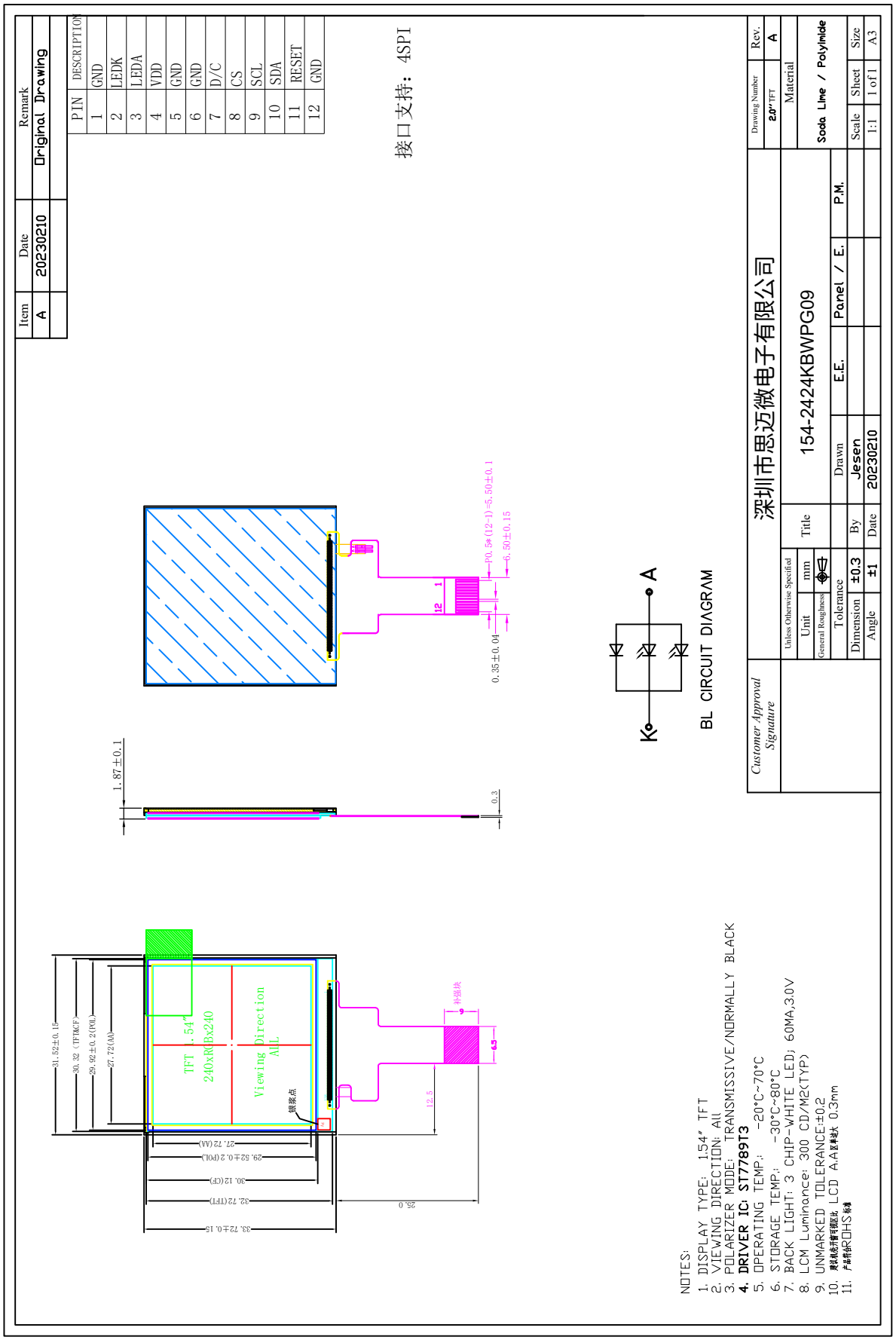
1.1 Description

154-2424KBWPG09 is a 240RGBX240 dot-matrix TFT LCD module. This module is composed of a TFT LCD Panel, driver ICs, FPC and a Backlight unit.

1.2 Features

NO.	Item	Contents	Unit
1	LCD Size	1.54	inch
2	Display Mode	Normally black	-
3	Resolution	240(H)RGB x240(V)	pixels
4	Pixel pitch	0.1155(H) x 0.1155(V)	mm
5	Active area	27.72(H) x 27.72(V)	mm
6	Module size	31.52(H) x 33.72(V) x1.9 (D)	mm
7	Pixel arrangement	RGB Vertical stripe	-
8	Interface	4 Line SPI	-
9	Display Colors	262K	colors
10	Drive IC	ST7789V3	-
11	Luminance(cd/m2)	400 (TYP)	Cd/m2
12	Viewing Direction	All View	Best image
13	Backlight	3 White LED Parallel	-
14	Operating Temp.	-20℃~ + 70℃	℃
15	Storage Temp.	-30℃~+ 80℃	℃
16	Weight	2.8	g

2. Mechanical Drawing

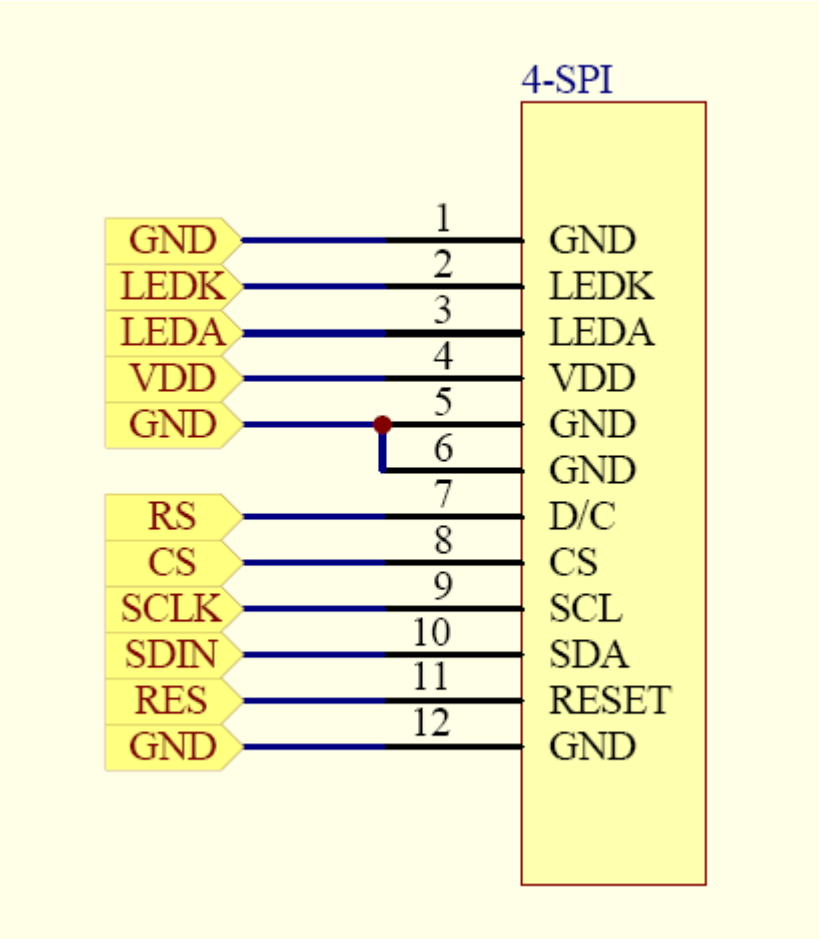


3. Pin Definition

FPC Connector is used for the module electronics interface.

NO.	Symbol	Description
1	GND	Power Ground
2	LEDK	LED Cathode
3	LEDA	LED Anode
4	VDD	Power Supply for Analog
5	GND	Power Ground
6	GND	Power Ground
7	D/C	Display data/command selection pin in 4-line serial interface.
8	CS	Chip selection pin ;Low enable ,high disable.
9	SCL	This pin is used to be serial interface clock
10	SDA	SPI interface input/output pin . the data is latched on the rising edge of the SCL signal.
11	RESET	This signal will reset the device and it must be applied to properly initialize the chip .Signal is active low.
12	GND	Power Ground

Note:



4. Electrical Characteristics

4.1 Absolute Maximum Ratings

Parameter	Symbol	Min	MAX	Unit	Notes
Supply Voltage (I/O)	VDD	-0.3	4.6	V	
Analog Supply Voltage	VDDIO	-0.3	4.6	V	
Logic Input Voltage	VIN	-0.3	VDDIO+0.5	V	
Operation Temperature	Top	-20	70	℃	
Storage Temperature	Tst	-30	80	℃	

4.2 Operating Conditions

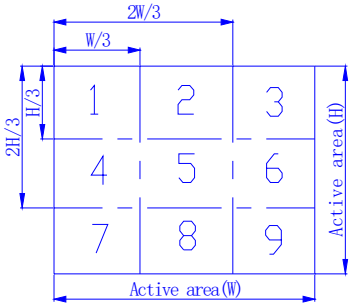
Parameter	Symbol	Min	TYP	MAX	Unit	Notes
System Voltage	VDD	2.5	2.8	3.3	V	
Gate Driver High Voltage	VGH	12.2	-	14.97	V	
Gate *Driver Low* Voltage	VGL	-12.5	-	-7.16	V	
Operating Current for V_{DD}	I_{DD}	-	8	10	mA	
Sleep_In Mode VDD	I_{dd}	-	15	30	uA	
Sleep_In Mode VDDIO	I_{ddio}	-	5	10	uA	

4.3 Backlight Unit

Parameter	Symbol	Min	TYP	MAX	Unit	Notes
Voltage for LED backlight	VLED	2.8	3.0	3.2	V	
Current for LED backlight	ILED	-	60	90	mA	3 LED
Power Consumption	Pbl	-	180	288	mW	1
Brightness	L_{br}	350	400	-	cd/m^2	2
LED Life time	-	20000	-	-	hr	3

Note:

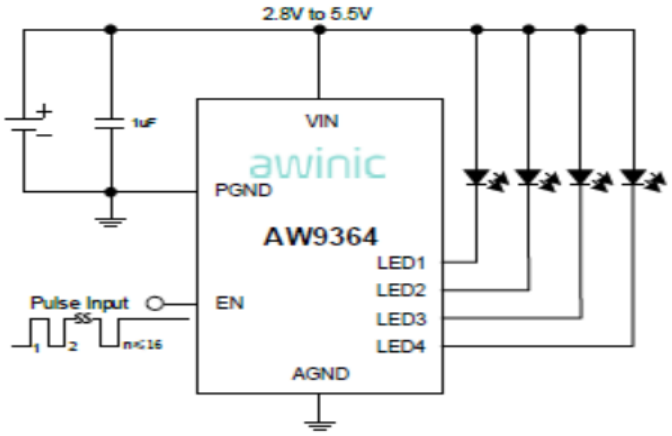
1. Where ILED =60mA , VLED=3.0V , Pbl= ILED x VLED
2. Uniform measure condition:
   a:Measure 9 point ,Measure location is show below:
   b:Uniform=(Min brightness/Max brightness)x100%
   c:Best Contrast.



3. The environmental conducted under ambient air flow ,at Ta=25±2°C,60%RH±5%

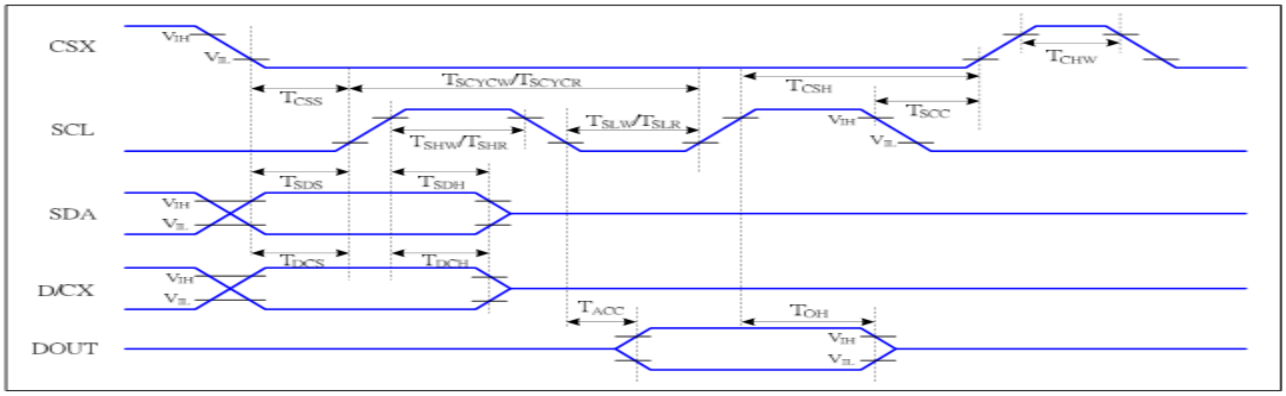
4.4 Backlight Recommended Circuit

Motherboard driver backlight is need constant current circuit , if threated voltage screen after light brightness difference . Current and power consumption of the machine are inconsistent , so recommend a backlight driving circuit is best rated current . It is recommended to use IC (AW9364) . The reference circuit is as follows:



4.5 AC Timing Characteristic of The LCD

Serial interface Characteristics(4-line serial):

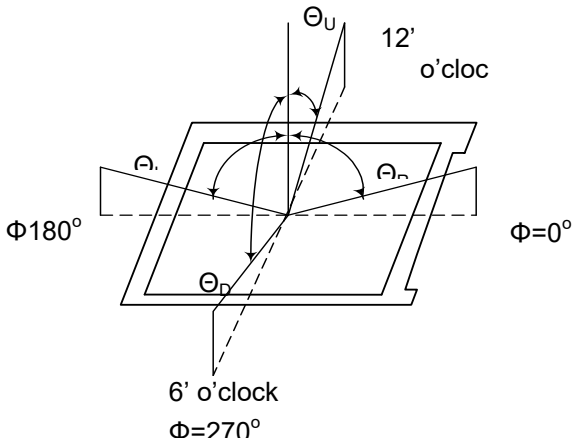


Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15		ns	
	T_{CSH}	Chip select hold time (write)	15		ns	
	T_{CSS}	Chip select setup time (read)	60		ns	
	T_{SCC}	Chip select hold time (read)	65		ns	
	T_{CHW}	Chip select “H” pulse width	40		ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	16		ns	-write command & data ram
	T_{SHW}	SCL “H” pulse width (Write)	7		ns	
	T_{SLW}	SCL “L” pulse width (Write)	7		ns	
	T_{SCYCR}	Serial clock cycle (Read)	150		ns	-read command & data ram
	T_{SHR}	SCL “H” pulse width (Read)	60		ns	
	T_{SLR}	SCL “L” pulse width (Read)	60		ns	
D/CX	T_{DCS}	D/CX setup time	10		ns	
	T_{DCH}	D/CX hold time	10		ns	
SDA (DIN)	T_{SDS}	Data setup time	7		ns	
	T_{SDH}	Data hold time	7		ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

5. OPTICAL CHARACTERISTICS

Item	Symbol	Measuring Conditions		Min.	Typ.	Max.	Unit	Remark
Viewing Angle	θ	ϕ = 0°	25 °C	70	80	-	Deg	Note1
		ϕ =180°	25 °C	70	80	-		
	θ	ϕ = 90°	25 °C	70	80	-		
		ϕ =270°	25 °C	70	80	-		
Brightness	L_{br}	--	-	350	400	-	Cd/m2	
Luminance Uniformity	ΔL	--	-	70	75	-		
Contrast Ratio	CR	--	25 °C	700	900	-	--	Note2
Response Time	Tr+Tf	θ = 0° ϕ = 0°	25 °C	-	30	40	ms	Note3
Color of CIE Coordinate	White	X	25 °C	-0.03	0.294	+0.03	--	BM-7A
		Y	25 °C		0.303			
	Red	X	25 °C		0.572			
		Y	25 °C		0.331			
	Green	X	25 °C		0.361			
		Y	25 °C		0.570			
	Blue	X	25 °C		0.154			
		Y	25 °C		0.092			
Transmittance (with polarizer)	--	--	--	4.4	4.9	-	%	--

Note 1 Definition of Viewing Angle:

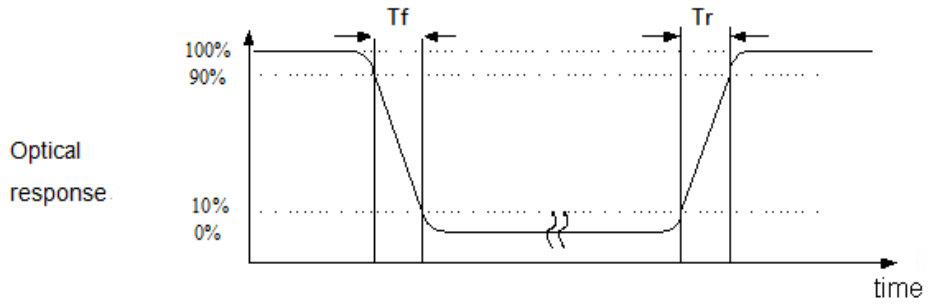


Note 2:Definition of Contrast Ratio (CR) :

measured at the center point of panel

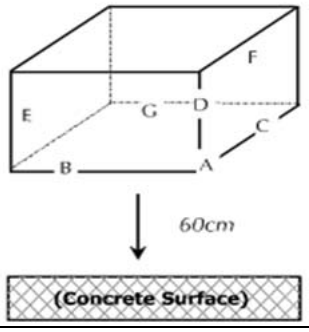
$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

Note 3: Definition of Response Time : Sum of Tr and Tf :



6. *Reliability*

Contents of Reliability Tests

No.	Item	Conditions	Note
1	High Temperature Operation	70°C±2°C, 120 hrs	
2	Low Temperature Operation	-20°C±2°C, 120 hrs	
3	High Temperature Storage	80°C±2°C, 120 hrs	
4	Low Temperature Storage	-30°C±2°C, 120 hrs	
5	High Temperature /Humidity Operation	60°C±2°C, 90% RH, 120 hrs	
6	Temperature Cycling	-10°C→25°C→60°C→25°C→-10°C 30min 5min 30min 5min 30min 10 cycle.	
7	Vibration Test	Total fixed amplitude:1.5mm. Vibration Frequerncy:10~55Hz One cycle 60 seconds to 3 direction of X,Y,Z each 15 minutes.	
8	ESD Test	Air Discharge:Apple ±4KV with 5 times.	
		Contact Discharge:Apple ±2KV with 5 times.	
9	Drop Test	To be measured after dropping from 60cm high on the concrete surface in packing state.  *Dropping method corner dropping:* *A corner: Once edge dropping.* *B, C, D edge: Once face dropping.* *E, F, G face: Once.*	

Note:

No charge on display and in operation under the following test condition.

Please note that the reliability test project requires the use of virgin samples

Condition : Unless otherwise specified ,tests will be conducted under the following condition.

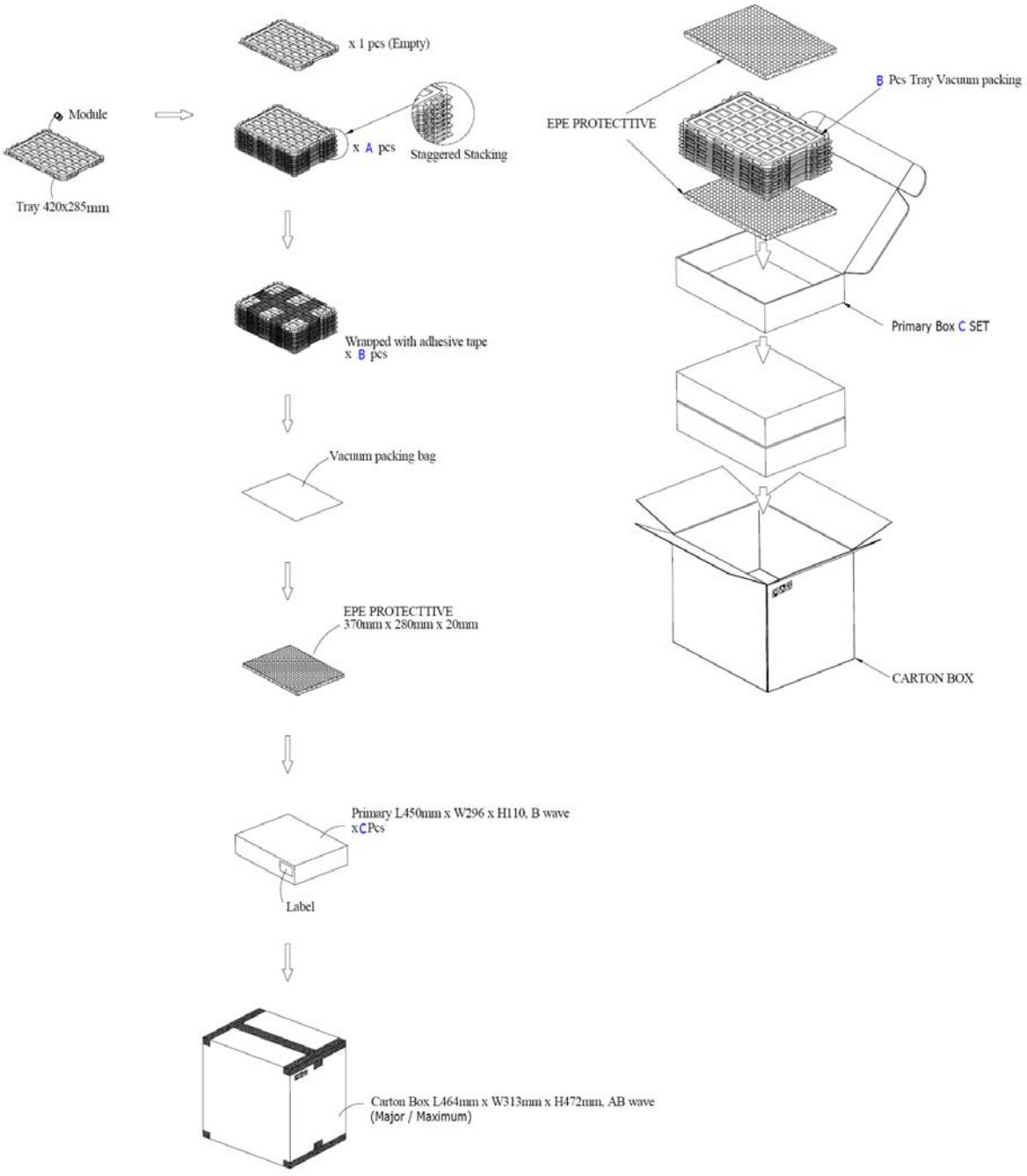
Temperature:20°C±5°C.

Humidity:65±5%RH.

Tests will be not conducted under functioning state.

7. Package Specifications

Item		Quantity	
Module		**720**	**per Primary Box**
Holding Trays	**(A)**	**15**	**per Primary Box**
Total Trays	**(B)**	**16**	**per Primary Box (Including 1 Empty Tray)**
Primary Box	**(C)**	**1~4**	**per Carton (4 as Major / Maximum)**



8. Incoming Inspection Standards

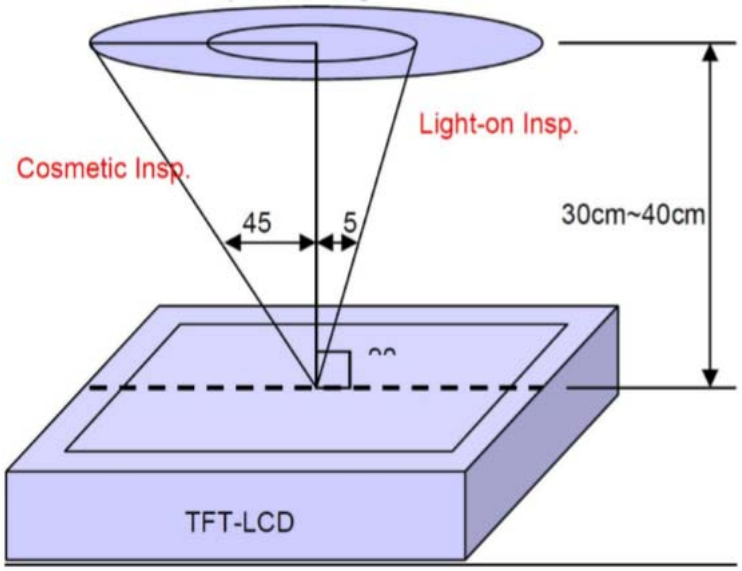
8.1. Inspection and Environment Conditions

8.1.1. Inspection Conditions:

(1) Inspection Distance :35 cm±5cm

(2) View Angle : Light-on Inspection Angle: ±5°

Cosmetic Inspection Angle: ±45°



(perpendicular to LCD panel surface)

8.1.2 Environment Conditions:

Ambient Temperature		**23**℃±5℃
Ambient Humidity		55±10％RH
Ambient Illumination	Cosmetic Inspection	More than 600 Lux
	Functional Inspection	300~500 Lux

8.1.3 Sampling Conditions:

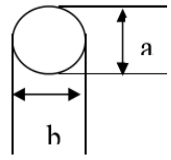
（1） Lot Size:Quantity of shipment lot per model

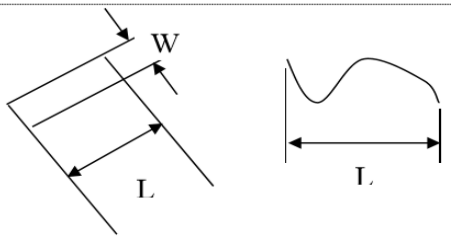
（2） Sampling Method:

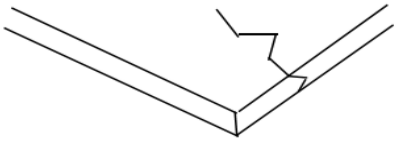
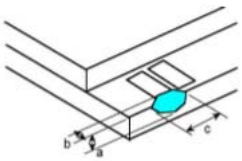
Sampling Plan		MIL-STD-105E
		Normal Inspection，Single Sampling
		Level II
AQL	Major Defect	0.65%
	Minor Defect	1.5%

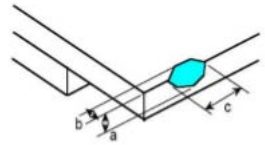
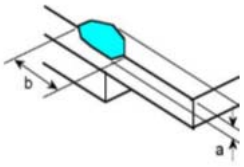
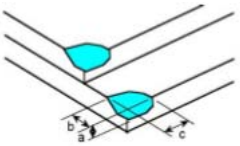
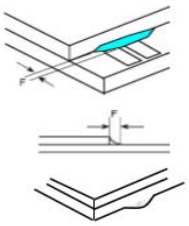
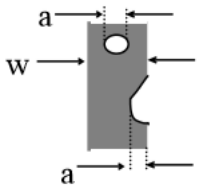
8.1.4 Inspection Criteria

8.1.4.1 Cosmetic Inspection(Panel):

<table>
<tr><th>Check Item</th><th>Classification</th><th>Criteria(Unit: mm)</th></tr>
<tr><td>Black / White spot
Foreign material
(Round type)
Pinholes Stain
Particles inside cell.</td><td>Minor</td><td> 

φ= (a + b) /2
<table>
<tr><th>Size \ Area</th><th>Acc. Qty</th></tr>
<tr><td>φ≤0.10</td><td>Ignore</td></tr>
<tr><td>0.10<φ≤0.15</td><td>2</td></tr>
<tr><td>0.15<φ≤0.20</td><td>1</td></tr>
<tr><td>0.20<φ</td><td>0</td></tr>
<tr><td>Total</td><td>2 no include φ≤ 0.10</td></tr>
</table>
Distance between 2 defects should more than 5mm apart.</td></tr>
<tr><td>Black and White line
Scratch
Foreign material
(Line type)</td><td>Minor</td><td> 

<table>
<tr><th>Length</th><th>Width</th><th>Acc. Qty</th></tr>
<tr><td>/</td><td>W ≦ 0.03</td><td>Ignore</td></tr>
<tr><td>L ≦ 2</td><td>0.03 < W ≦ 0.05</td><td>1</td></tr>
<tr><td>/</td><td>0.05 < W</td><td>0</td></tr>
<tr><td colspan="2">Total</td><td>1</td></tr>
</table>
Distance between 2 defects should more than 5mm apart.
Scratches not viewable through the back of the display are acceptable.</td></tr>
<tr><td>Glass Crack</td><td>Minor</td><td> 
LCD with extensible crack line is unacceptable(When press the cracked LCD Area, the line will expand,we define it is extensible crack line)</td></tr>
<tr><td>Glass Chipping Pad Area
  </td><td>Minor</td><td>
<table>
<tr><th>Length and Width</th><th>Acc. Qty</th></tr>
<tr><td>c < 5.0, b< 0.4</td><td>Ignore</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Check Item</th><th>Classification</th><th>Criteria(Unit: mm)</th></tr>
<tr><td>Glass Chipping Rear
Of Pad Area
  </td><td>Minor</td><td><table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>c > 3.0, b< 1.0</td><td>1</td></tr><tr><td>c< 3.0, b< 1.0</td><td>2</td></tr><tr><td>c< 3.0, b< 0.5</td><td>4</td></tr><tr><td colspan="2">a<Glass Thickness</td></tr></table></td></tr>
<tr><td>Glass Chipping
Except Pad Area
  </td><td>Minor</td><td><table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>c ≤0.6, b< 5.0</td><td>Ignore</td></tr><tr><td colspan="2">a<Glass Thickness</td></tr></table></td></tr>
<tr><td>Glass Corner Chipping
  </td><td>Minor</td><td><table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>c < 2.0, b< 1.5</td><td>Ignore</td></tr><tr><td>c < 1.5, b< 2</td><td>Ignore</td></tr><tr><td colspan="2">a<Glass Thickness</td></tr></table></td></tr>
<tr><td>Glass Burr
  </td><td>Minor</td><td>Glass burr don’t affect assemble and module dimension.<table><tr><th>Length</th><th>Acc. Qty</th></tr><tr><td>F < 0.5</td><td>Ignore</td></tr></table></td></tr>
<tr><td>FPC Defect
  </td><td>Minor</td><td>1.Dent , pinhole width a＜W/2.
(W:circuitry width)
2.Open circuit is unacceptable.
3.No oxidation, contamination and distortion.</td></tr>
<tr><td>Bubble on Polarizer</td><td>Minor</td><td><table><tr><th>Diameter</th><th>Acc. Qty</th></tr><tr><td>φ≤0.15</td><td>Ignore</td></tr><tr><td>0.15 <φ≤0.20</td><td>2</td></tr><tr><td>0.20 <φ≤0.30</td><td>1</td></tr><tr><td>0.3 < φ</td><td>None</td></tr></table></td></tr>
</table>

<table>
<tr><th>Check Item</th><th>Classification</th><th>Criteria(Unit: mm)</th></tr>
<tr><td>Dent on Polarizer</td><td>Minor</td><td>
<table>
<tr><td>Diameter</td><td>Acc. Qty</td></tr>
<tr><td>φ≤0.15</td><td>Ignore</td></tr>
<tr><td>0.15 <φ≤0.20</td><td>2</td></tr>
<tr><td>0.20 <φ≤0.30</td><td>1</td></tr>
<tr><td>0.3 < φ</td><td>None</td></tr>
</table>
</td></tr>
<tr><td>Screen deformation
 
</td><td>/</td><td>Test for insertion of plug gauge at highest warping point:
H≤0.25mm
The client has special requirements, according to drawing.</td></tr>
<tr><td>Bezel</td><td>/</td><td>1.No rust, distortion on the Bezel.
2.No visible fingerprints, stains or other contamination.</td></tr>
<tr><td>Touch Panel</td><td>/</td><td>D:Diameter W: width L: length
1.Spot: D≤0.2 is acceptable
0.2＜D≤0.3, acceptable Inspection and Environment Conditions
2dots are acceptable and the distance between defects
Should more than 5mm.
D＞0.3 is unacceptable
2.Dent: D＞0.3 is unacceptable.
3.Scratch: W≤0.03,L≤10 is acceptable,
0.03＜W≤0.1, L≤10, acceptable Inspection and Environment Conditions
Distance between 2 defects should more than 5 mm.
W＞0.1 is unacceptable.</td></tr>
<tr><td>PCB</td><td>/</td><td>1.No distortion or contamination on PCB terminals.
2.All components on PCB must same as documented on the BOM/component layout.
3.Follow IPC-A-600F.</td></tr>
<tr><td>Soldering</td><td>/</td><td>Follow IPC-A-610C standard.</td></tr>
<tr><td>Leak</td><td>/</td><td>Yellow light, OK。White light, According to the limit sample</td></tr>
</table>

Check Item	Classification	Criteria(Unit: mm)
Electrical Defect	Major	The below defects must be rejected. 1. Missing vertical / horizontal segment. 2. Abnormal Display. 3. No function or no display. 4. Current exceeds product specifications. 5. LCD viewing angle defect. 6. No Backlight. 7. Dark Backlight. 8. Touch Panel no function. 9. Dark Dot –one Allowed. 10. Bright Dot- one Allowed.

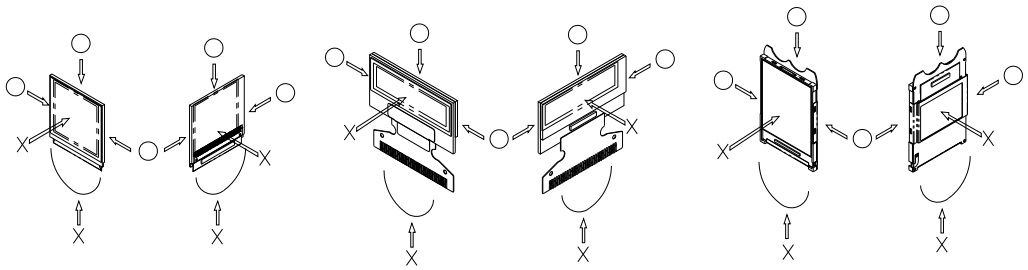
9. Precautions When Using These TFT Display Modules

9.1 Handling Precautions

1) Since the display panel is being made of glass, do not apply mechanical impacts such us dropping from a high position.
2) If the display panel is broken by some accident and the internal organic substance leaks out, be careful not to inhale nor lick the organic substance.
3) If pressure is applied to the display surface or its neighborhood of the TFT display module, the cell structure may be damaged and be careful not to apply pressure to these sections.
4) The polarizer covering the surface of the TFT display module is soft and easily scratched. Please be careful when handling the TFT display module.
5) When the surface of the polarizer of the TFT display module has soil, clean the surface. It takes advantage of by using following adhesion tape.
   * Scotch Mending Tape No. 810 or an equivalent

   Never try to breathe upon the soiled surface nor wipe the surface using cloth containing solvent such as ethyl alcohol, since the surface of the polarizer will become cloudy.

   Also, pay attention that the following liquid and solvent may spoil the polarizer:
   * Water
   * Ketone
   * Aromatic Solvents
6) Hold TFT display module very carefully when placing TFT display module into the system housing. Do not apply excessive stress or pressure to TFT display module. And, do not over bend the film with electrode pattern layouts. These stresses will influence the display performance. Also, secure sufficient rigidity for the outer cases.



7) Do not apply stress to the driver IC and the surrounding molded sections.
8) Do not disassemble nor modify the TFT display module.
9) Do not apply input signals while the logic power is off.